

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – INDUSTRY PROBLEM SOLVING TASK

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO4 – Precision (BL4)	Assessment:	Industry Problem solving task
Weightage:	30%	Total Marks:	100
CO4 Statement:	Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Instructions to Students

- Identify the relevant concepts of cache hierarchy, SRAM, DRAM, MRAM, NAND Flash, RRAM, memory latency, bandwidth, and virtual memory paging.
- Analyze the memory-access requirements of smartphones, cloud servers, autonomous vehicles, edge AI devices, and gaming consoles, considering latency, bandwidth, capacity, and power consumption.
- Apply suitable memory-hierarchy techniques such as cache optimization, appropriate memory technology selection, data locality, paging optimization, and hierarchical memory organization.
- Compute performance measures such as memory access time, latency, bandwidth, throughput, speedup, hit/miss rates, and resource or power utilization where applicable.
- Interpret the results by evaluating performance improvements, memory bottlenecks, technology trade-offs, and the suitability of the proposed memory hierarchy for each application.

QUESTIONS

1. A high-frequency financial trading system must process large volumes of market data with extremely low response time. Analyze the roles of CPU caches, DRAM, and persistent storage in handling frequently accessed trading data. Considering latency, data locality, capacity, and bandwidth, propose a memory hierarchy that minimizes data-access delays and supports high transaction throughput.
2. A medical imaging workstation processes large CT and MRI datasets while simultaneously running image-enhancement and visualization applications. Evaluate the interaction between cache memory, DRAM, and high-capacity storage during image processing. Propose an optimized memory hierarchy that balances access latency, memory capacity, bandwidth, and cost to improve overall processing performance.
3. A smart surveillance system receives continuous video streams from multiple high-resolution cameras and must detect objects in real time. Analyze the memory requirements for storing video frames, intermediate processing data, and frequently accessed detection parameters. Compare cache, DRAM, and non-volatile memory technologies and design a hierarchy that minimizes memory bottlenecks while maintaining high data throughput.
4. A scientific computing system performs large-scale simulations involving matrices and continuously accessed numerical datasets. Analyze how cache capacity, data locality, cache misses, DRAM bandwidth, and memory latency affect the execution of computational kernels. Recommend memory-hierarchy improvements that can reduce cache misses, increase effective memory bandwidth, and improve overall simulation performance.
5. A wearable health-monitoring device continuously collects sensor readings while operating under strict battery constraints. Analyze the trade-offs among SRAM, DRAM, and non-volatile memory in terms of latency, capacity, energy consumption, and data retention. Design a memory hierarchy that provides rapid access to frequently processed sensor data while minimizing power consumption and extending battery life.

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Weightage(%)	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Marks Evaluation:

Q. No	Understanding of Industry Problem (20)	Application of Engineering Concepts (25)	Solution Design (25)	Use of Tools (15)	Analysis (10)	Professionalism (5)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/ 30	/ 20	/ 15	/ 10		/ 100

Course Coordinator

Faculty Incharge

1. High-Frequency Financial Trading System

A high-frequency trading system processes a very large amount of market data and requires extremely low response time. The memory hierarchy must therefore provide low latency, high bandwidth, and sufficient capacity.

Role of CPU Cache

CPU cache is the fastest memory after registers. It stores frequently accessed data such as:

- Current stock prices
- Order-book information

- Trading parameters
- Frequently used algorithm variables

L1 cache is the fastest but smallest, while L2 and L3 provide larger capacity with slightly higher latency. Keeping frequently accessed data in cache reduces the number of DRAM accesses.

Role of DRAM

DRAM provides much larger capacity than cache and is used for active market datasets, order books, temporary calculations, and running applications. Although slower than cache, high-speed DRAM provides sufficient capacity and bandwidth for large workloads.

Role of Persistent Storage

NVMe SSDs provide large-capacity persistent storage. They are suitable for:

- Historical market data
- Transaction logs
- Backups
- System configuration

Persistent storage should not be used for latency-critical trading operations because it is much slower than cache and DRAM.

Proposed Memory Hierarchy

Registers → L1 Cache → L2 Cache → L3 Cache → High-Speed DRAM → NVMe SSD

Frequently accessed data should be kept close to the CPU. Data locality, prefetching, and reducing cache misses can further improve performance.

Conclusion

This hierarchy provides **very low latency, high bandwidth, and high transaction throughput** by keeping critical trading data in cache, active data in DRAM, and historical data in persistent storage.

2. Medical Imaging Workstation

Medical imaging workstations process large CT and MRI datasets while performing image enhancement, reconstruction, segmentation, and visualization. Therefore, both memory capacity and bandwidth are important.

Role of Cache Memory

Cache stores frequently accessed image-processing data, instructions, and intermediate calculations. Since image processing often accesses neighboring pixels repeatedly, good spatial and temporal locality can improve cache performance.

Role of DRAM

DRAM acts as the main working memory. It stores:

- CT/MRI image volumes
- Image buffers
- Intermediate processing results
- Segmentation data
- Visualization information

A large DRAM capacity prevents frequent swapping to storage.

Role of High-Capacity Storage

NVMe SSDs store complete CT/MRI datasets, patient studies, processed images, and application files. SSDs are much faster than traditional hard disks and allow large datasets to be loaded quickly.

Proposed Memory Hierarchy

CPU/GPU Registers → L1/L2/L3 Cache → Large DRAM → NVMe SSD → Archival Storage

Optimization

- Use image **tiling/blocking** to improve cache locality.
- Use high-bandwidth DRAM for large image transfers.
- Use GPU memory for parallel image processing.
- Use NVMe SSDs for fast dataset loading.
- Avoid unnecessary movement of image data between memory levels.

Conclusion

Cache provides **low latency**, DRAM provides **large working capacity and bandwidth**, and SSD provides **high-capacity storage**. This combination gives good performance at reasonable cost.

3. Smart Surveillance System

A smart surveillance system receives continuous video streams from multiple high-resolution cameras and performs real-time object detection. It must process frames quickly without creating memory bottlenecks.

Memory Requirements

The system requires memory for:

- Incoming video frames
- Multiple frame buffers
- Intermediate feature maps
- Object-detection results
- Neural-network parameters
- Tracking information

Cache Memory

Cache or SRAM stores frequently accessed detection parameters, small data structures, and processing variables. Its very low latency is useful for real-time processing.

DRAM

DRAM stores current video frames, frame buffers, intermediate results, and other large working datasets. High DRAM bandwidth is important because multiple cameras generate continuous data.

Non-Volatile Memory

Flash or NVMe storage is used for recorded video, logs, configuration files, and historical information. It provides large capacity and retains data without power.

Proposed Hierarchy

CPU/GPU Registers → Cache → DRAM/GPU Memory → NVMe/Flash Storage

Performance Improvements

1. Use multiple frame buffers so capturing and processing can happen simultaneously.
2. Use DMA to transfer frames efficiently.
3. Use GPU processing for parallel object detection.
4. Keep frequently used parameters in cache.
5. Compress older video before storing it.

6. Avoid unnecessary copying of video frames.

Conclusion

A combination of **fast cache, high-bandwidth DRAM/GPU memory, and high-capacity non-volatile storage** provides low latency and high throughput for real-time surveillance.

4. Scientific Computing System

Scientific computing systems perform large-scale simulations involving matrices, vectors, and numerical datasets. Memory performance strongly affects the execution time of computational kernels.

Cache Capacity

If the frequently used working set fits into cache, the CPU can access it quickly. If the dataset is larger than the cache, frequent cache misses occur, causing additional DRAM accesses.

Data Locality

Two types of locality are important:

- **Temporal locality:** Recently accessed data is likely to be reused.
- **Spatial locality:** Data near the accessed location is likely to be used soon.

Good locality reduces cache misses and improves performance.

Cache Misses

When data is not available in cache, it must be obtained from lower memory levels. A large number of cache misses increases memory latency and reduces CPU utilization.

DRAM Bandwidth and Latency

Large scientific calculations require continuous data transfer. If DRAM bandwidth is insufficient, the processor may remain idle while waiting for data. High memory latency also reduces performance, especially for irregular memory accesses.

Recommended Improvements

1. Use **blocking/tiling** for matrix operations.
2. Optimize data structures for sequential access.
3. Use hardware/software prefetching.
4. Increase DRAM bandwidth using multiple memory channels.

5. Reduce unnecessary data movement.
6. Use SIMD/vector processing.
7. Optimize NUMA memory placement in multi-processor systems.

Proposed Hierarchy

Registers → L1/L2/L3 Cache → High-Bandwidth DRAM → SSD

Conclusion

Improving data locality, increasing cache reuse, reducing cache misses, and providing sufficient DRAM bandwidth can significantly improve the performance of large-scale scientific simulations.

5. Wearable Health-Monitoring Device

A wearable health-monitoring device continuously collects sensor readings while operating under strict battery constraints. The memory hierarchy must therefore balance speed, capacity, energy consumption, and data retention.

SRAM

SRAM is very fast and consumes relatively little power for small working sets. It is suitable for:

- Latest sensor readings
- Temporary buffers
- Real-time calculations
- Frequently accessed variables

However, SRAM is expensive and has limited capacity.

DRAM

DRAM provides more capacity than SRAM and can store larger temporary datasets. However, DRAM requires refresh operations, which consume additional energy.

Therefore, DRAM should be used only when the workload requires additional working memory.

Non-Volatile Memory

Flash or other non-volatile memory retains data even when power is removed. It is suitable for:

- Historical sensor data
- Firmware

- Configuration information
- Activity records

It has higher latency than SRAM and DRAM, so it should not be used for frequently accessed real-time data.

Proposed Memory Hierarchy

CPU Registers → SRAM/Cache → Low-Power DRAM → Flash

Frequently processed sensor data should remain in SRAM. Larger temporary datasets can be placed in DRAM, while long-term information should be stored in Flash.

Power Optimization

- Keep frequently accessed data in SRAM.
- Reduce unnecessary DRAM access.
- Put unused memory into low-power mode.
- Batch sensor readings before writing to Flash.
- Compress data to reduce storage requirements.
- Avoid continuous Flash writes.

Conclusion

The proposed hierarchy provides **fast access to important sensor data while minimizing power consumption**. SRAM handles real-time processing, DRAM provides additional working capacity, and non-volatile memory provides long-term data retention.